Jhih-Jhong (Johnny) Lin

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Education

M.S. in Computer Science and Information Engineering Aug. 2026 – Present
National Taiwan University *Taipei, Taiwan*

B.S. in Computer Science and Information Engineering Aug. 2022 – Jul. 2026
National Taiwan University *Taipei, Taiwan*

- **Cumulative GPA**: 4.23/4.30 | **Class Rank**: 7/150
- **Relevant Courses**: Algorithm Design and Analysis (A+), Quantum Algorithms (A+), Cryptography and Network Security (A+), Theoretical Aspects of Modern Cryptography (A+), Post-Quantum Cryptography (A-), Programming Language Theories and Type Systems (A+), Compiler Design (A+), Intuitionistic Logic (A+), Formal Methods (A), Formal Languages and Automata Theory (A+), Intro to Abstract Algebra I (A-), Operating Systems (A+), Virtual Machines (A)

Research Interests

Formal verification, program analysis, cryptography.

Research Experience

Student Researcher Sept. 2024 – Present
Network Security Lab (NSLab), CSIE, National Taiwan University *Taipei, Taiwan*
Advisor: Prof. Hsu-Chun Hsiao

- Investigated integration of directed fuzzers with recurring vulnerability detection tools.
- Explore formal methods, especially on verification of constant-time property for cryptographic programs.

Research Intern / Project Assistant Jul. 2025 – Sept. 2026
Institute of Information Science (IIS), Academia Sinica *Taipei, Taiwan*
Advisor: Prof. Bow-Yaw Wang

- Extended CryptoLine using OCaml to model floating-point instructions and verify them with the Bitwuzla SMT solver.
- Developed abstract interpretation techniques for verification of floating-point cryptographic programs.

Visiting Research Student Jun. 2026 – Aug. 2026
Department of Informatics, University of California, Irvine *Irvine, California*
Advisor: Prof. Joshua Garcia

- Surveyed 16 agentic LLM security benchmarks and experimentally evaluated AgentDojo, AgentHarm, and ASB for use in cybersecurity education.
- Designed a reproducible evaluation methodology extending BetterBench with education-specific criteria and analyzed benchmark behavior across locally hosted LLMs.

Teaching Experience

Teaching Assistant, Introduction to Computational Logic Fall 2025
Department of Computer Science and Information Engineering, National Taiwan University *Taipei, Taiwan*

- Evaluated biweekly homework assignments for a class of 20+ students.

Teaching Assistant, Network Administration and System Administration Spring 2025
Department of Computer Science and Information Engineering, National Taiwan University *Taipei, Taiwan*

- Co-designed homework and exam problems for a class of 50+ students, providing hands-on experience with shell-scripting and computer security concepts.

Selected Projects

Toward Verification of Floating-Point Computation in Falcon by Abstract Interpretation
Research Project at Academia Sinica, advised by Prof. Bow-Yaw Wang
Code (OCaml) Project Report (Interim)

Grey-Box Symbolic Execution with Likely-Invariant Generation in CPAchecker
Formal Methods Course Project, advised by Prof. Nian-Ze Lee
Code (Java) Project Report

Evaluating Agentic LLM Security Benchmarks for Cybersecurity Education: A Reproducible Study
Research Project at UC Irvine (UCInspire Program), advised by Prof. Joshua Garcia
Artifact Slides

An Evaluation on Possibilities of RCE and Defense Mechanism in Electron-Based Applications
Cryptography and Network Security Course Project
Co-authored with: Kang-Yu Fan, Tsan-Huan Hsu, Yi-Ting Lo Project Report

CTF Challenge Design: Cryptography, Fuzzing, and DNS Cache Poisoning
Developed as a TA for the Network Administration and System Administration course
DNS Cache Poisoning Cryptography and Fuzzing

MerkleDir.sh: A shell script that creates Merkle trees from directories and generates/verifies inclusion proofs
Developed as a TA for the Network Administration and System Administration course
Code (Bash)

Terminal-Based P2P and Centralized Chatroom Implementation
Computer Networks Course Project
Co-authored with: Wen-Chun Kao Code (C++)

Work Experience

Product Development Intern Sept. 2026 – Present
Synology, Inc. *New Taipei, Taiwan*
- Automate the conversion of 100+ manual QA test specifications into executable tests with LLM agents.

Network Administrator (Member of NASA Team) Sept. 2023 – Jun. 2025
Department of Computer Science and Information Engineering, National Taiwan University *Taipei, Taiwan*
- Maintained Cisco and Juniper switch infrastructure serving 600+ faculty members and students.
- Built automated monitoring and backup workflows using Prometheus and Grafana for newly deployed switches.

Awards

Dean's List, NTU, Fall 2023

Dean's List, NTU, Fall 2022

Skills

Languages: Mandarin (native); English (advanced, TOEFL iBT 5.5/6)

Programming: C, Python, OCaml, Bash; working knowledge of C++, Java, and Haskell

Software: Linux, Git, LaTeX, Docker, CPAchecker, CryptoLine, Compiler (flex, yacc), Fuzzing (AFL++)